Trading Volatility with VIX, VIX3M, and eVRP

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Editorial English title. English translation of the Japanese post; performance claims are those of the author.

Source: https://x.com/kashin_moneylab/status/2119440436670738737

English translation

I regularly look through research papers, and I would like to briefly introduce a strategy that trades volatility using VIX and VIX3M as its main signals and has delivered quite good backtest performance.

First, annualize the S&P 500's volatility over the past 10 days, then subtract it from the current VIX to calculate an indicator called eVRP.

Next, compare VIX with VIX3M: if VIX < VIX3M, the market is considered calm; if VIX > VIX3M, it is considered to be in a state of alert.

Combine eVRP with this VIX assessment to decide whether to bet on volatility falling or rising. See the attached images for the detailed signals.

In practice, the trades use ETNs such as VXX, entering either short or long positions.

Had this strategy been used since 2008, its total return would have exceeded +1,200%, growing assets to more than 13 times their initial value, substantially outperforming the eightfold growth of the S&P 500 including dividends.

Furthermore, even including the global financial crisis, the maximum drawdown was held to around 30%, with a Sharpe ratio of 1.0.

It is also shown that adding just a 10–20% allocation to this strategy alongside an S&P 500 buy-and-hold portfolio improves the overall Sharpe ratio by about 20%.

Please see the paper for details.

Linked paper: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5316487

Original Japanese text

も普段から論文を漁っているのだが、実はVIXとVIX3Mを主要シグナルにして、ボラティリティを売買する戦略がかなり良いパフォーマンスをバックテストで残しているので簡潔に紹介する。

まずS&P500の過去10日間におけるボラティリティを年率換算し、それを現在のVIXから差し引いてeVRPという指標を算出する。

続いてVIXとVIX3Mを比較し、VIX < VIX3Mなら市場は平穏、VIX > VIX3Mなら警戒状態と判断する。

そしてeVRPとVIX判定を組み合わせてボラティリティが下がる方に賭けるのか、上がる方に賭けるのかという売買の意思決定を行うというものである（詳細のシグナル添付画像参照）

実際の売買ではVXXといったETNを利用し、空売りもしくは買いでエントリーを行う。

この戦略を2008年から運用すればトータルリターンは+1,200%超えで資産が13倍を超える成績であり、S&P500（配当込み）の資産8倍を大きく上回る。

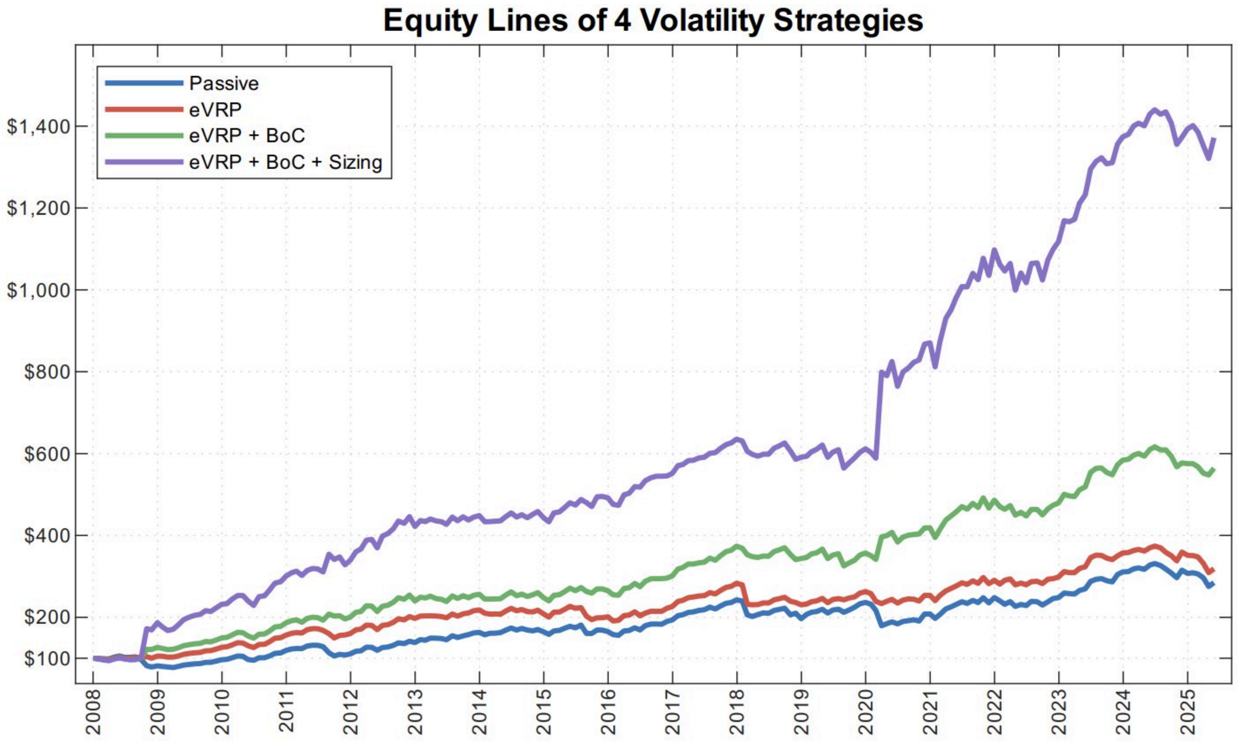
それだけではなく、リーマンショックを含めても最大下落率は30%程度に抑制されおり、シャープレシオは1.0をつけている。

そしてS&P500のバイアンドホールドに10~20%というわずかな割合でこの戦略を混ぜるだけで、全体のシャープレシオが約20%向上することも示されている。

詳細は論文をぜひご覧いただきたい。

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5316487

Original attachment 1



Original attachment 2

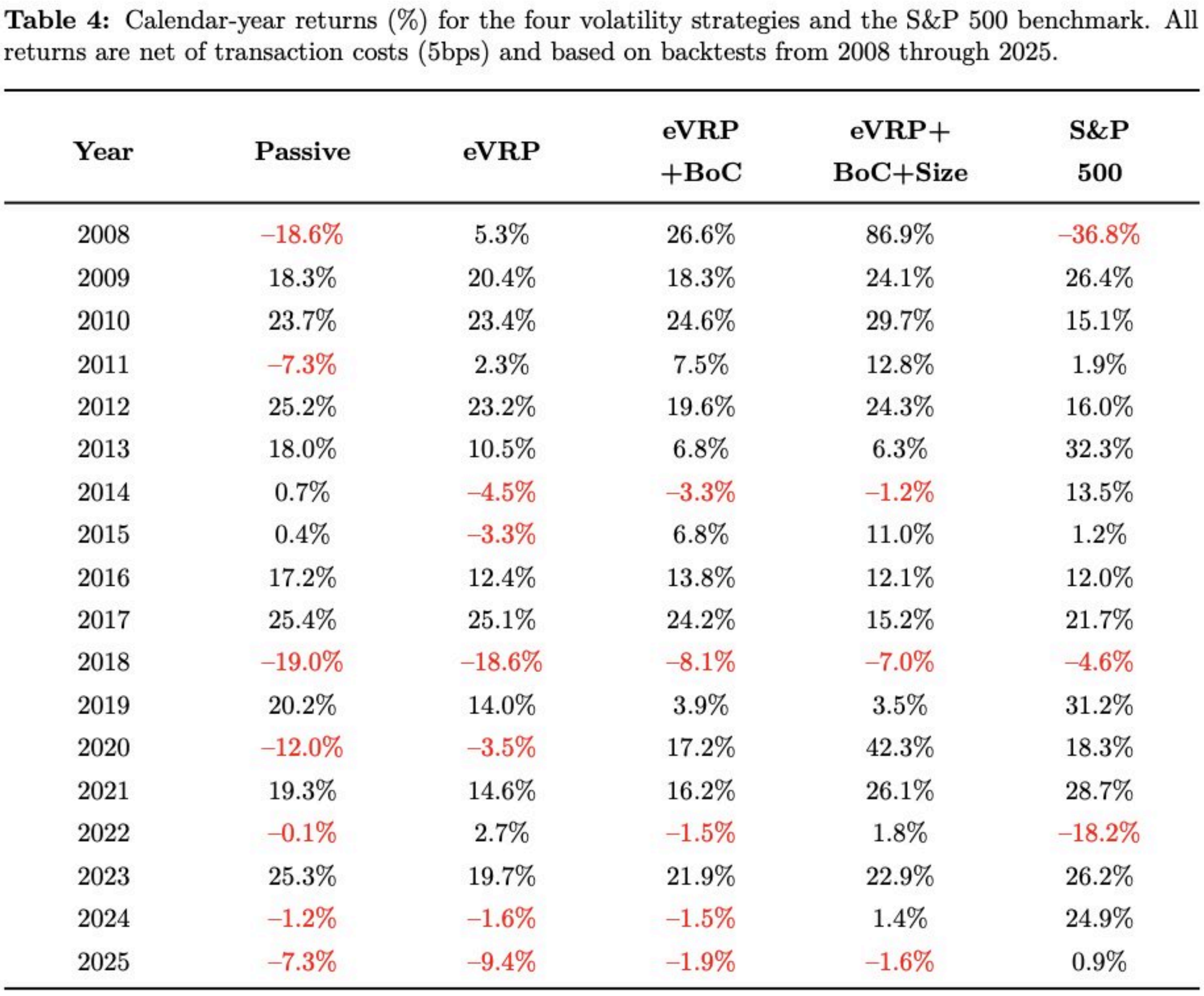


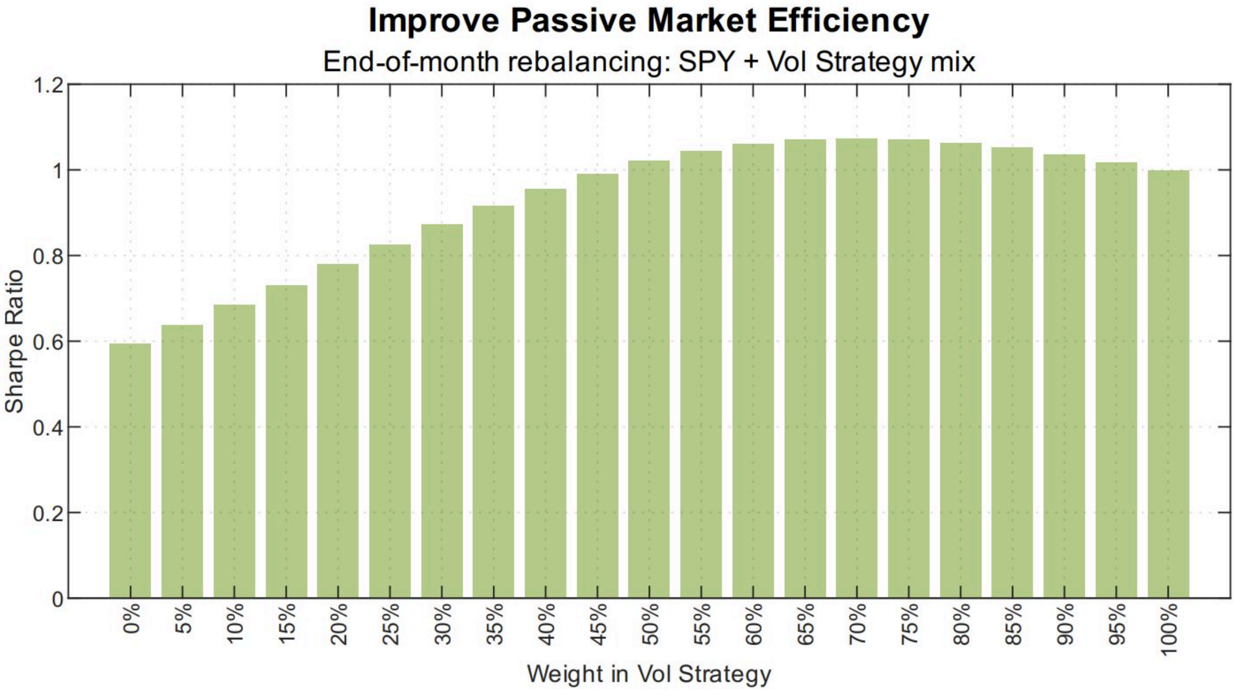
Table 4: Calendar-year returns (%) for the four volatility strategies and the S&P 500 benchmark. All returns are net of transaction costs (5bps) and based on backtests from 2008 through 2025.

Year	Passive	eVRP	eVRP +BoC	eVRP+ BoC+Size	S&P 500
2008	–18.6%	5.3%	26.6%	86.9%	–36.8%
2009	18.3%	20.4%	18.3%	24.1%	26.4%
2010	23.7%	23.4%	24.6%	29.7%	15.1%
2011	–7.3%	2.3%	7.5%	12.8%	1.9%
2012	25.2%	23.2%	19.6%	24.3%	16.0%
2013	18.0%	10.5%	6.8%	6.3%	32.3%
2014	0.7%	–4.5%	–3.3%	–1.2%	13.5%
2015	0.4%	–3.3%	6.8%	11.0%	1.2%
2016	17.2%	12.4%	13.8%	12.1%	12.0%
2017	25.4%	25.1%	24.2%	15.2%	21.7%
2018	–19.0%	–18.6%	–8.1%	–7.0%	–4.6%
2019	20.2%	14.0%	3.9%	3.5%	31.2%
2020	–12.0%	–3.5%	17.2%	42.3%	18.3%
2021	19.3%	14.6%	16.2%	26.1%	28.7%
2022	–0.1%	2.7%	–1.5%	1.8%	–18.2%
2023	25.3%	19.7%	21.9%	22.9%	26.2%
2024	–1.2%	–1.6%	–1.5%	1.4%	24.9%
2025	–7.3%	–9.4%	–1.9%	–1.6%	0.9%

Original attachment 3

Table 2: Summary of the volatility trading strategies.

	Strategy	**Positioning Rule**	**Size**	**Rebal.**	**TCost**	**Signal**	**Order**
1	**Passive**	VIXSHORT	20%	±2%	5 bps	3:45pm	MOC
2	**eVRP**	VIXSHORT if eVRP > 0	20%	±2%	5 bps	3:45pm	MOC
		Cash otherwise					
3	**eVRP + BoC**	VIXSHORT if eVRP > 0, VIX $<$ VIX3M	20%	±2%	5 bps	3:45pm	MOC
		VIXSHORT if eVRP ≤ 0, VIX $<$ VIX3M	10%				
		VIXLONG if eVRP ≤ 0, VIX $>$ VIX3M	20%				
		Cash otherwise					
4	**eVRP + BoC + Sizing**	VIXSHORT if eVRP > 0, VIX $<$ VIX3M	VIX%	±2%	5 bps	3:45pm	MOC
		VIXSHORT if eVRP ≤ 0, VIX $<$ VIX3M	0.5×VIX%				
		VIXLONG if eVRP ≤ 0, VIX $>$ VIX3M	VIX%				
		Cash otherwise					

Original attachment 4